Quant summer internship talk

Siddhant, Tanish, Shiven, Gautam
2023 batch

Quant roles offered

Quant researcher

Quant developer

Quant trader

Quant analyst

....lot more off-campus

Speakers

Quant researcher

Quant developer

Quant trader

Quant analyst

....lot more off-campus

Speakers

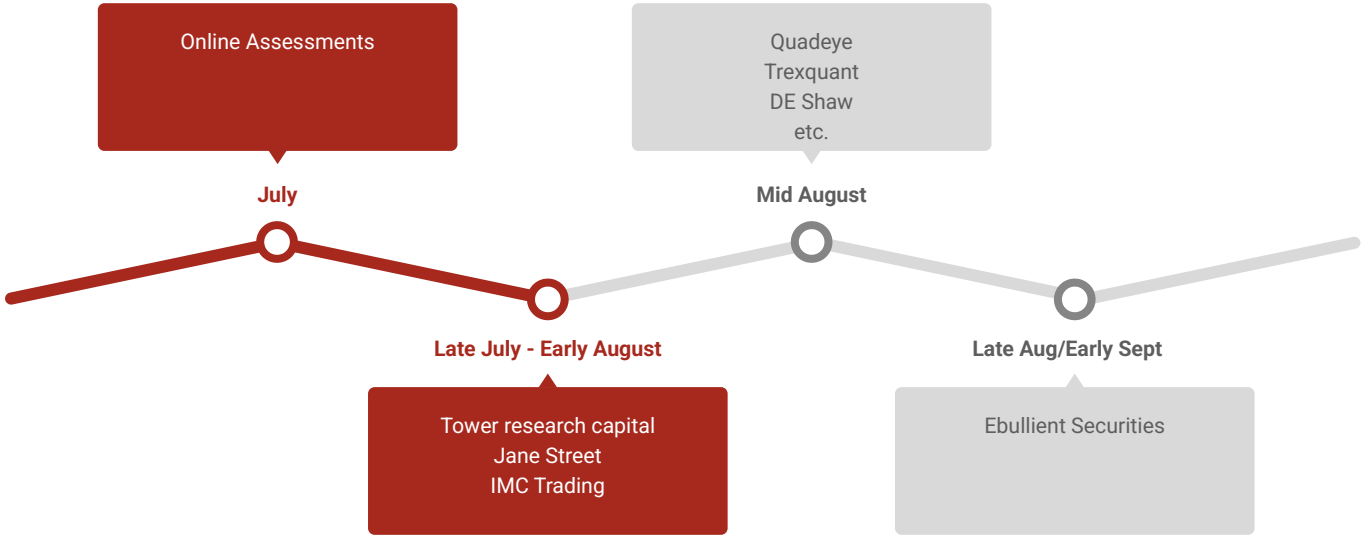
Siddhant Attavar
Quant Research & Dev
Tower Research Capital

Tanish Jain
Software Engineering
IMC Trading

Shiven Sharma
Systems Engineering
Quadeye

Gautam Singhvie
Quantitative Trader
IMC Trading

Timeline


Online Assessments
July
Quadeye
Trexquant
DE Shaw
etc.
Mid August
Late July - Early August
Tower research capital
Jane Street
IMC Trading
Late Aug/Early Sept
Ebullient Securities

Disclaimer

Most of this talk is based on speculation and is just a collection of our personal experiences while preparing for Quant SI roles.

Do not follow this as a roadmap, but rather as a blueprint of what to expect while preparing.

Quant Summer Internship Talk:

Siddhant Attavar (2023A7)
Incoming Quantitative Research and Development Intern
Tower Research Capital, North Moore team
London Office, UK

Application Process:

- Eligibility
  - Branch: CS only
  - CGPA (Pre-PS1): 9+
- Online Assessment
  - 3 Coding Questions, 1 hour
  - 12 MCQs (CS fundamentals)
  - Previous years: Debugging questions
- Interviews
  - 3 technical interviews
  - Clear communication is important
  - Explain your though process while solving

Competitions:

- International Collegiate Programming Contest
- Olympiads (Linguistics, Informatics)
- Competitive Programming platforms
  - Codeforces: codeforces.com
  - CodeChef: codechef.com
  - AtCoder: atcoder.jp
  - CSES: cses.fi/problemset

Internships and Projects:

- Internships
  - PS-1
  - Off campus internships
- Projects
  - SOP/independent research
  - Tech clubs
  - Personal projects
- Quality > Quantity
- Should demonstrate high technical expertise in a field

Resoures:

- Competitive Programming
- C++: learncpp.com
- Math and puzzles:
  - Brainsteller: brainstellar.com
  - 50 challenging problems in probability

My personal (biased) advice:

Graduating in 2028

- Brush up acads (CS CDCs + PnS)
- Study 1 programming language well (C++/Python/Java)
- DSA for interviews
- Give mock interviews to improve communication

Graduating in 2029/30:

- Explore 1 technical skill deeply
- Participate in competitions
- Build a portfolio of projects
- Look out for off-campus opportunities

Questions?

IMC Trading: Software

TANISH JAIN

2023 AA

Eligibility (for SWE Intern)

- Branch: CS, ECE, EEE, ENI
- CGPA: 8.0

Interview Process

- Online Assessment
- Group Discussion
- Design Interview
- Coding Interview
- Behavioural Interview

Some numbers

- ~ 36 people cleared OA and took part in GD
- From the collective pool, ~10 people were shortlisted for SWE Interviews
- Another 3* for trading

- Selections: 3 for SWE, 1 for trading

What helped me

Disclaimer:

These are the things that I did, not with any focus on landing a quant SW internship, but just because I liked doing it. The things that worked out for me may not work out for you, and there are many things I chose not to do, that may be beneficial for your learning and growth.

Please use this as a reference, not as a roadmap.

DSA

- Most important part of the selection process; OA and interviews depended almost entirely on DSA knowledge.
- What I used for learning/practice:
  - Striver's A2Z sheet
  - Whatever questions I found fun on Leetcode
  - Occasionally Codeforces
- What to focus on:
  - Getting sufficient practice in all important topics, including but not limited to: DP, Graphs, Arrays, Strings, Trees, etc.
  - Solving questions in time. Roughly <15 mins for LC Easy, 30 mins for LC Medium and 45 mins - 1 hour for LC Hard
- What not to focus on:
  - Quantity of questions solved
  - Memorizing questions and approaches

AI/ML

- Was not directly asked in the interviews, but was a major part of my resume.
- Club projects
- Stanford CS 229 (https://youtube.com/playlist?list=PLoROMvodv4rMiGQp3WXShtMGgzqpfVfbU&si=n2tjzcPlniY3cr4z)
- Stanford CS 231n (https://youtube.com/playlist?list=PL3FW7Lu3i5JvHM8ljYj-zLfQRF3EO8sYv&si=zw1qCF1IA7Wmrh1H)
- Machine Learning Specialization on Coursera

Internships/Projects

- Club projects
- SOP projects
- Personal projects
- PS1
- Off Campus Internships

General tips

- Balancing DSA, projects, CGPA
- Mock OAs and Interviews
- Programming languages

Questions

Quadeye

Roles

- Quant
- Systems Engineer
- Product (Web Dev)

Eligibility

- Branches: CS only
- CGPA Criteria: 8.5+

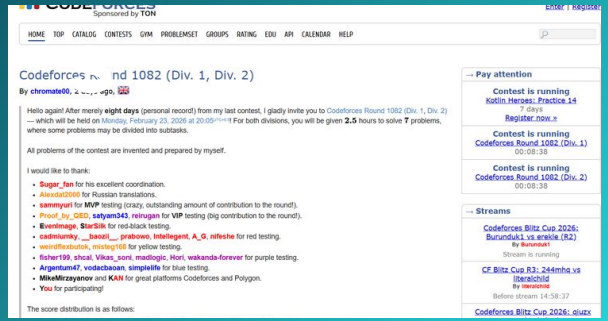
What the company looks for

- Algorithms
- OOPs
- CPP
- Probability Puzzles
- Brain Teasers
- Algorithmic Efficiencies
- Quick Thinking

Resources

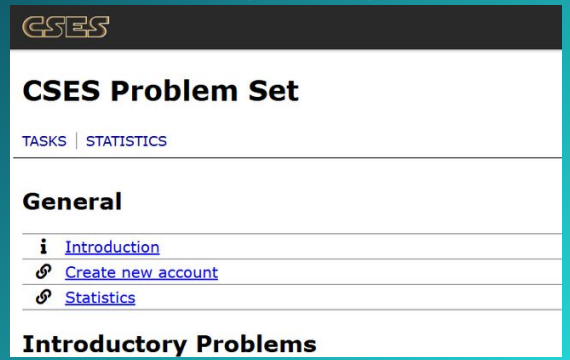
Some sites which helped me

Codeforce



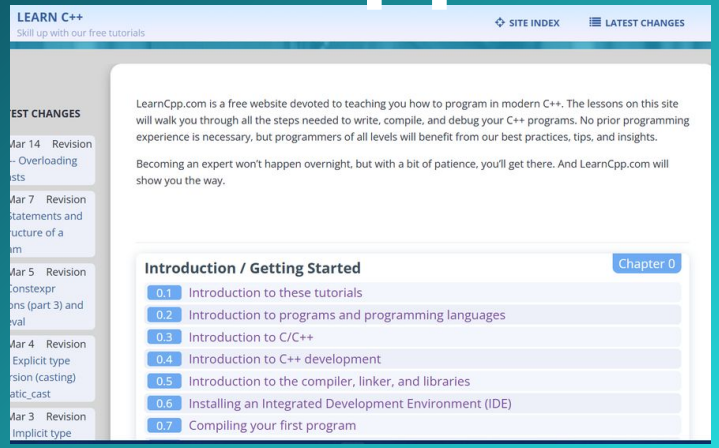
Leetcode

CSES

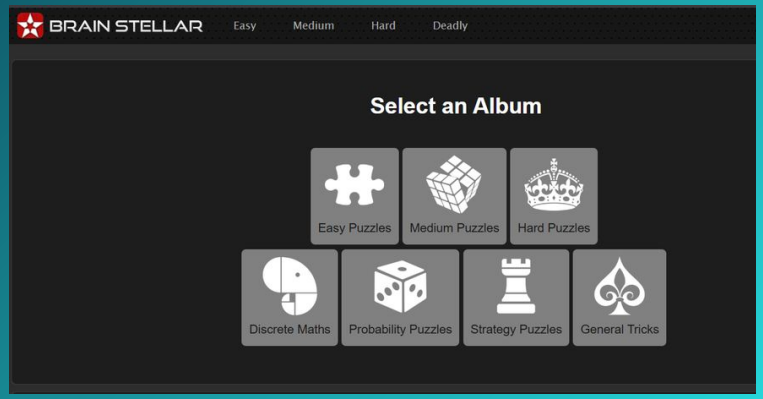


Resources

Learncpp.com



Brainstellar



General Preparation Tips

Try to build up speed while solving problems

Keep fundamentals strong

OOP based questions are there in pretty much every interview.

Stamina

You need to be prepared for solving a lot of questions in a short span of time.

Strong cpp knowledge is also essential in some quant dev roles

General Preparation Tips

Balancing Acads with prep

Try to minimise your time wasted studying extra for acads. focus on whats important.

Thinking out loud during interview is a major plus point

Projects

Be prepared with some projects which could be helpful as a talking point

You can try building existing projects but improving their efficiency

Interview Style

Fast Paced Multi round

Each interview lasts for a short time and covers one or two problems.

Be honest with your interviewer

Mix of questions

Probability questions,

DSA questions,

Brain teasers

Talking about how you approach a problem is very beneficial

Questions?

Quant Summer Internship Talk

Gautam Singhvie ('23 CSE)

Trading Intern Role

Official requirements:

1. 8 CGPA
2. NO Branch Requirements
3. Competency in a programming lang(Cpp, Java or Python)

What they look for:

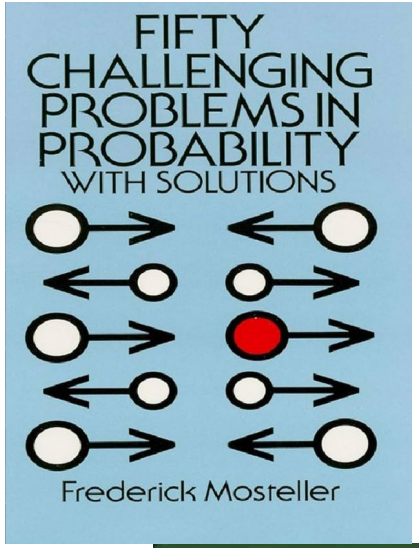
1. Excellent Modelling Skills
2. An intuitive feel for probability and math
3. Top - notch problem solving and improvisation
4. The willingness to take on new challenges

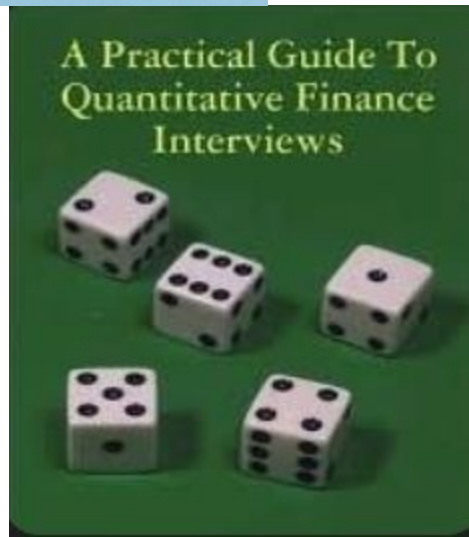
How it showed in the process

1. A math focussed OA along with programming problems
2. A dynamic market making GD round
3. 2 Technical Interviews
4. 1 Hr Interview

How to build these skills?

1. Some can be gained from books:
   a. 50 Challenging Problems in Probability
   b. Green Book (Intuition comes from the dark?)
   c. + Many other equivalent resources
   d. - Puzzle sites (Endgame stuff)
2. For most of the rest - Competitions!!
   a. As many as you can get your hands on!
   b. IMC's Prosperity, Optiver's Integral Cup,
   c. Citadel's Terminal, Trex quant Hangman
   d. ICPC, IICPC Quantfest and Codefest, etc
   e. ++Competitive Programming![Long Term]





Other Imp Stuff

1. Acads, and how to not go HAM!
2. Projects, and the inspiration to make something good
3. OA and Interview prep, and how to skip the “first trap”
4. Motivation, and how to not get burnout
5. First years, and the long road
6. Second years, and is there even time for this?

Questions?

Contact for details

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